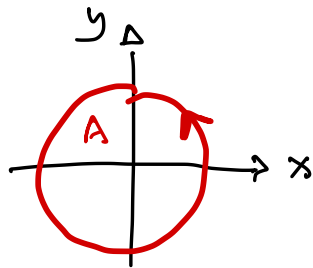


2.4.3 Examples

01/10/2026

Example 1: Verify Green's theorem for $A = \{ (x,y) \in \mathbb{R}^2 : x^2 + y^2 < 1 \}$

and $F(x,y) = (y^2, x)$. $\text{dom}(F) = \mathbb{R}^2$.



polar coord.

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$dx dy = r dr d\theta$$

$$1) \iint_A \text{curl} F dx dy = \iint_A (1 - 2y) dx dy$$
$$\text{curl} F(x,y) = 1 - 2y$$

$$= \int_0^1 \int_0^{2\pi} (1 - 2r \sin \theta) r dr d\theta$$

$$= \int_0^1 \left(r \int_0^{2\pi} d\theta - 2r^2 \int_0^{2\pi} \sin \theta d\theta \right) dr = \int_0^1 r \int_0^{2\pi} d\theta dr$$

~~$\int_0^{2\pi} \sin \theta d\theta = 0$~~

$$= \underline{\underline{\pi}}$$

$$2) \quad \partial A = \{ (x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1 \}$$

Positive oriented param:

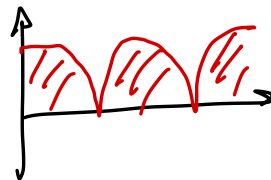
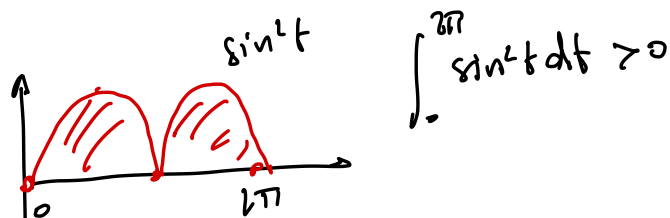
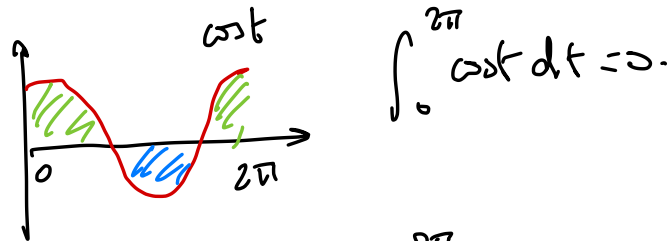
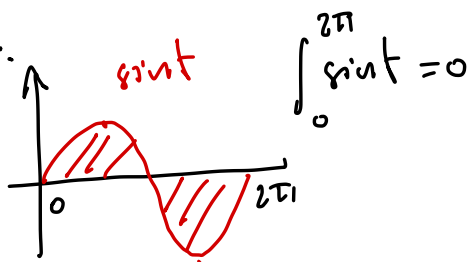
$$\gamma: [0, 2\pi] \rightarrow \mathbb{R}^2$$

$$t \mapsto \gamma(t) = (\cos t, \sin t), \quad \gamma'(t) = (-\sin t, \cos t)$$

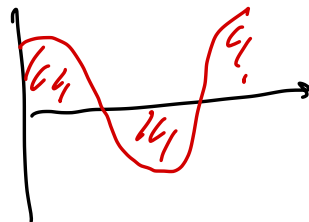
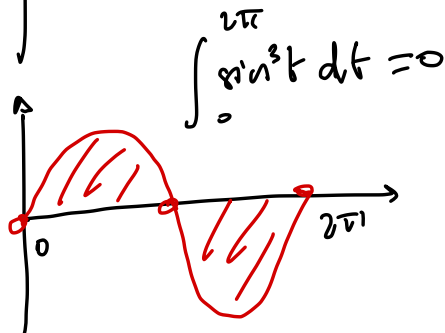
$$\int_{\partial A} F \cdot d\mathbf{l} = \int_0^{2\pi} F(\gamma(t)) \cdot \gamma'(t) dt = \int_0^{2\pi} (\sin^2 t, \cos t) \cdot (-\sin t, \cos t) dt$$

$$= - \underbrace{\int_0^{2\pi} \sin^3 t dt}_{=0. \text{ (PS)}} + \underbrace{\int_0^{2\pi} \cos^2 t dt}_{\neq 0} = \pi = \iint_A \text{curl } F dx dy \quad \checkmark$$

Note:



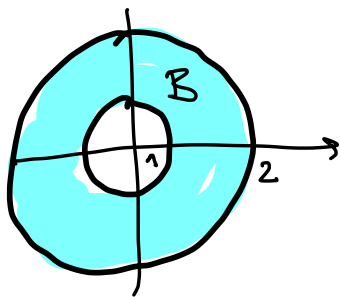
$\int_0^{2\pi} \cos^2 t \, dt > 0$



$\int_0^{2\pi} \cos^3 t \, dt = 0$

...

Example 2: Verify Green's theorem for $B = \{ (x,y) \in \mathbb{R}^2 : 1 < x^2 + y^2 < 4 \}$.
and $F(x,y) = (x^2y, 2xy)$



$$1) \iint_A \text{curl} F \, dx \, dy = \iint_A (2y - x^2) \, dx \, dy$$

$$\stackrel{\substack{\text{polar} \\ \text{coordinates}}}{=} \int_1^2 \int_0^{2\pi} (2r \sin \theta - r^2 \cos^2 \theta) \, r \, d\theta \, dr$$

$$= \int_1^2 \left(2r^2 \underbrace{\int_0^{2\pi} \sin \theta \, d\theta}_{=0} - r^3 \underbrace{\int_0^{2\pi} \cos^2 \theta \, d\theta}_{=\pi} \right) \, dr$$

$$= -\pi \int_1^2 r^3 \, dr = -\frac{15}{4} \pi$$

$$2) \int_{\partial B} F \cdot dl, \quad \partial B = \Gamma_0 \cup \Gamma_1$$

$$\gamma_0: [0, 2\pi] \rightarrow \mathbb{R}^2$$

$$t \mapsto \gamma_0(t) = 2(\cos t, \sin t)$$

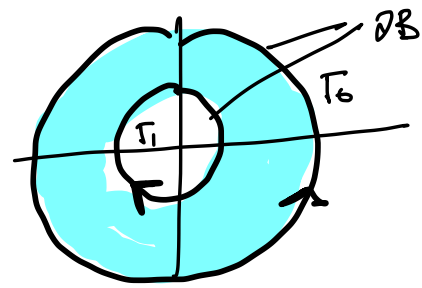
$$\gamma_1: [0, 2\pi] \rightarrow \mathbb{R}^2$$

$$t \mapsto \gamma_1(t) = (\cos t, -\sin t)$$

$$\int_{\partial B} F \cdot dl = \int_{\Gamma_0} F \cdot dl + \int_{\Gamma_1} F \cdot dl$$

$$\int_{\Gamma_0} F \cdot dl = \int_0^{2\pi} F(\gamma_0(t)) \cdot \gamma_0'(t) dt$$

$$= \int_0^{2\pi} (8 \cos^4 t \sin t, 8 \cos t \sin t) \cdot (-2 \sin t, 2 \cos t) dt$$



$$= 16 \int_0^{2\pi} \underbrace{(-\omega^2 t \sin^2 t)}_{-\pi/4} + \underbrace{\omega^2 t \sin t}_{=0} dt = -4\pi$$

$$\int_{\Gamma_1} F \cdot d\mathbf{l} = \int_0^{2\pi} F(x_1(t)) \cdot x_1'(t) dt$$

$$= \int_0^{2\pi} (\dots , \dots) \cdot (-\sin t, -\omega t) dt$$

$$= \underbrace{\int_0^{2\pi} \omega^2 t \sin^2 t dt}_{=\pi/4} + 2 \underbrace{\int_0^{2\pi} \omega^2 t \sin t dt}_{=0} = \pi/4$$

$$\int_{\partial B} F \cdot d\mathbf{l} = \int_{\Gamma_0} F \cdot d\mathbf{l} + \int_{\Gamma_1} F \cdot d\mathbf{l} = -4\pi + \pi/4 = -\frac{15}{4} \pi.$$

$$= \iint_B \operatorname{curl} F \, dx \, dy \quad \checkmark$$

2.4.5 Corollaries of Green's Theorem

• Corollary 1: Divergence theorem in $\mathbb{R}^2$

Let $A \subset \mathbb{R}^2$ be a regular domain whose boundary ∂A is positively oriented, let $\nu: \partial A \rightarrow \mathbb{R}^2$ the field of outer unit normal vectors on ∂A .

Let $F: \bar{A} \rightarrow \mathbb{R}^2$ be a vector field s.t. $F \in C^1(\bar{A}, \mathbb{R}^2)$

Then :

$$\iint_A \operatorname{div} F(x,y) \, dx \, dy = \int_{\partial A} F \cdot \nu \, dl$$

or :

$$\iint_A \left(\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} \right) dx \, dy = \int_{\partial A} (F_1 \nu_1 + F_2 \nu_2) \, dl.$$

Exercise (home):

Derive div. theorem starting from Green's theorem:

Tricks:

- write ψ as a function of x'

$$\rightarrow \text{curl } \phi = \frac{\partial \phi_2}{\partial x} - \frac{\partial \phi_1}{\partial y} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} = \text{div } F$$